

MEDAN BELAWAN, Indonesia

WMO: 960330

Lat: 3.7883N

Lon: 98.7147E

Elev: 3

StdP: 101.29

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	23.3	23.8	21.1	15.7	27.5	21.8	16.4	26.9	8.1	27.9	6.5	28.6	1.0	250	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	5.7	32.9	26.8	32.4	26.7	32.0	26.6	27.8	31.4	27.5	31.2	27.2	30.9	3.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.3	30.4	26.3	21.8	30.0	26.1	21.5	29.8	88.9	31.7	87.1	31.4	85.9	31.2	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 7.7	6.3	5.3	22.1	36.0	1.0	2.6	21.4	37.9	20.9	39.4	20.4	40.8	19.7	42.7		
(5)			20.9	29.1	1.0	0.7	20.2	29.6	19.7	30.0	19.1	30.4	18.4	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	27.4	27.8	28.4	28.6	28.5	28.3	28.0	27.7	27.5	27.5	27.3	(6)	
(7)		DBStd	0.98	0.89	0.79	0.86	0.91	0.90	0.93	0.90	0.92	0.87	0.79	0.70	0.86	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6559	539	498	569	558	576	557	567	559	532	542	525	537	(10)
(11)		CDD18.3	3518	281	265	311	308	317	307	309	301	282	283	275	279	(11)
(12)		CDH23.3	39829	2971	2964	3682	3754	3856	3728	3626	3435	3062	2997	2879	2876	(12)
(13)		CDH26.7	14150	878	988	1418	1514	1554	1529	1406	1258	1041	927	836	802	(13)
(14)	Wind	WSAvg	1.7	1.5	1.8	1.9	1.9	1.8	1.8	1.7	1.7	1.6	1.5	1.5	(14)	
(15)	Precipitation	PrecAvg	2553	159	95	147	143	214	174	183	235	307	339	277	280	(15)
(16)		PrecMax	3791	394	319	417	332	437	400	354	446	594	587	557	557	(16)
(17)		PrecMin	1672	71	36	30	27	90	78	56	95	165	166	93	94	(17)
(18)		PrecStd	562	79	60	101	82	95	73	77	88	101	95	95	123	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	32.1	32.7	33.2	33.3	33.2	33.6	32.9	32.2	32.0	31.5	31.5	(19)
MCWB			26.4	26.4	26.9	27.2	26.9	26.2	26.2	26.4	26.2	26.2	26.0	25.9	(20)	
2%		DB	31.0	31.5	32.2	32.5	32.5	32.4	32.2	32.1	31.6	31.2	30.8	30.9	(21)	
		MCWB	26.0	26.1	26.6	27.0	26.9	26.6	26.5	26.6	26.3	26.1	26.0	25.9	(22)	
5%		DB	30.5	31.0	31.7	32.0	32.0	31.9	31.7	31.5	31.2	30.7	30.4	30.4	(23)	
		MCWB	25.8	25.9	26.5	26.8	26.9	26.6	26.4	26.5	26.2	26.0	25.9	25.8	(24)	
10%		DB	30.0	30.5	31.2	31.4	31.5	31.5	31.2	31.0	30.6	30.2	30.0	29.8	(25)	
		MCWB	25.7	25.8	26.2	26.5	26.7	26.5	26.3	26.3	26.1	25.9	25.9	25.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	27.1	27.7	28.0	28.1	28.0	28.1	29.1	27.8	27.1	26.7	26.9	(27)
MCDWB			30.4	30.8	31.5	31.9	32.0	31.5	32.0	31.2	30.9	30.3	30.0	30.2	(28)	
2%		WB	26.6	26.7	27.2	27.5	27.6	27.5	27.2	27.3	26.9	26.6	26.6	26.5	(29)	
		MCDWB	30.0	30.3	31.1	31.4	31.5	31.1	31.0	30.6	30.6	30.0	29.8	29.7	(30)	
5%		WB	26.3	26.4	26.8	27.1	27.2	27.1	26.9	26.8	26.6	26.3	26.2	26.2	(31)	
		MCDWB	29.5	29.9	30.6	31.0	31.0	30.9	30.6	30.4	30.2	29.7	29.4	29.3	(32)	
10%		WB	26.0	26.1	26.5	26.8	27.0	26.8	26.6	26.5	26.2	26.1	26.1	25.9	(33)	
		MCDWB	29.2	29.6	30.3	30.6	30.7	30.5	30.3	30.1	29.8	29.5	29.3	29.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.2	5.5	5.8	5.7	5.8	5.9	6.0	5.8	5.7	5.3	5.0	4.9	(35)
MCDBR			5.8	6.0	6.2	6.2	6.3	6.3	6.7	6.4	6.5	5.9	5.6	5.6	(36)	
MCWBR			2.4	2.4	2.6	2.6	2.8	3.0	3.1	3.1	2.9	2.6	2.4	2.3	(37)	
5% WB		MCDWB	5.5	5.7	5.9	5.8	5.9	6.0	6.2	6.0	6.1	5.5	5.2	5.2	(38)	
		MCWBR	2.5	2.6	2.7	2.7	2.9	3.1	3.2	3.2	3.0	2.6	2.4	2.4	(39)	
		Clear-Sky Solar Irradiance	taub	0.477	0.521	0.504	0.473	0.468	0.480	0.471	0.489	0.488	0.495	0.452	0.454	(40)
taud	2.265		2.117	2.184	2.280	2.313	2.279	2.308	2.241	2.246	2.220	2.352	2.347	(41)		
Ebn at Noon	850		819	832	841	825	803	815	817	833	831	866	864	(42)		
Edh at Noon	140		165	155	138	129	132	129	141	143	147	127	127	(43)		
(44)	All-Sky Solar Radiation	RadAvg	4.43	4.92	5.12	4.86	4.77	4.86	4.73	4.73	4.53	4.34	4.09	4.05	(44)	
RadStd		0.26	0.27	0.29	0.22	0.28	0.28	0.25	0.33	0.36	0.48	0.19	0.28	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		+0.19	+0.11	N/A	N/A	+0.40	+0.56	N/A	N/A	N/A	N/A

Nomenclature: See separate page